JEE Mathematics

Detailed JEE Main + Advanced Syllabus

Exam Coverage	JEE Main + JEE Advanced aligned topic map
Format	Module-wise syllabus with detailed bullet-wise subtopics
Use Case	Foundation planning, course brochure, chapter scheduling and curriculum design

Structured Syllabus

Unit 1: Sets, Relations and Functions

- Sets and their representation; subsets, power set, universal set, Venn diagrams, union, intersection, complement and Cartesian product.
- Relations and functions: ordered pairs, domain, codomain and range; types of relations; one-one, onto, many-one functions; composition and inverse of functions.
- Elementary understanding of algebraic, trigonometric, exponential, logarithmic and greatest integer functions; graphs and transformations where relevant.

Unit 2: Complex Numbers and Quadratic Equations

- Complex numbers in the form a + ib, modulus, argument, conjugate and Argand plane representation.
- Algebra of complex numbers, polar form basics, triangle inequality and geometric interpretation.
- Quadratic equations in real and complex domain; nature of roots, relation between roots and coefficients, formation of quadratic equations.

Unit 3: Sequences, Series and Binomial Expansion

- Arithmetic progression, geometric progression and harmonic progression; nth term, sum of finite terms and infinite GP.
- Means between numbers; insertion of arithmetic and geometric means; applications in algebraic problems.
- Binomial theorem for positive integral index, general term, middle term, simple identities and approximations.
- Permutations and combinations: fundamental principle of counting, factorial notation, circular arrangements, selections and distributions.

Unit 4: Matrices and Determinants

- Types of matrices, equality, addition, multiplication, transpose and elementary operations.
- Determinants of order up to 3, minors, cofactors, properties of determinants and area interpretation.
- Adjoint and inverse of a matrix; solving simultaneous linear equations using matrix method and Cramer’s rule.

Unit 5: Trigonometry

- Trigonometric ratios, identities, reduction formulae and standard transformations.
- Trigonometric equations, properties of triangles, sine rule, cosine rule, projection formula and area of triangle.
- Inverse trigonometric functions, principal values, properties and standard manipulations used in JEE problems.

Unit 6: Coordinate Geometry

- Straight lines: slope, intercepts, equation in different forms, angle between lines, distance of a point from a line and family of lines.
- Circles: standard and general equation, tangent, normal, radical axis and common tangents.
- Conic sections: parabola, ellipse and hyperbola — standard equations, eccentricity, latus rectum, parametric representation and tangent-normal basics.

Unit 7: Differential Calculus

- Limits, continuity and differentiability of algebraic, trigonometric, logarithmic and exponential functions.
- Methods of differentiation: chain rule, product rule, quotient rule, implicit differentiation and logarithmic differentiation.
- Applications of derivatives: tangents and normals, monotonicity, maxima and minima, rate of change and approximation.

Unit 8: Integral Calculus and Differential Equations

- Indefinite integrals and standard methods including substitution, partial fractions and integration by parts.
- Definite integrals and their properties; area bounded by simple curves and coordinate axes.
- Differential equations of first order and first degree; variable separable form and applications.

Unit 9: Vectors, 3D Geometry and Probability

- Vectors in two and three dimensions, magnitude, direction cosines, scalar product, vector product and scalar triple product.
- 3D geometry: line and plane, angle between lines and planes, shortest distance concepts in basic form.
- Probability, conditional probability, Bayes theorem, binomial distribution basics, mean and variance; elementary statistics measures.